

$$T_p \frac{dy}{dt} + y = K_p f(t), \quad f(0) = 0$$

$$H(s) = \frac{Y(s)}{F(s)}$$

$$\Rightarrow T_p (s Y(s) - y(0)) + Y(s) = K_p F(s)$$

$$\Rightarrow \frac{Y(s)}{F(s)} = \frac{K_p}{T_p s + 1}$$